

SCOPE OF WORK: QUOTATION REQUEST

Digital D.O. Tracking & Invoice Verification

Solvex by AI Playground Labs | Version 1.0 | May 2026

CLIENT
[Solvex: Confidential]

INDUSTRY
Civil Engineering /
Construction

LOCATION
Singapore

STATUS
Awaiting Quotation

PART A: PROJECT BRIEF

1. Executive Summary

Build a mobile-first web application that digitizes manual Delivery Order (D.O.) handling process. Drivers capture D.O.s via mobile phone camera; AI extracts the data automatically; supervisors verify daily; finance reconciles against supplier invoices instantly. Replaces the current paper + Excel workflow that loses documents, delays month-end, and exposes the company to overpayment risk.

A working prototype has already been built to validate the concept. We are looking for a developer to build the production version. Your job is to rebuild it as a production system with proper backend, auth, and data persistence.

2. Background

The client is a SG civil engineering firm with government contracts. They operate 25+ trucks, 28+ drivers, and process approximately 262 delivery orders per month from 3 material suppliers.

2.1 Current Pain Points

- D.O. paper copies frequently lost between site and office
- No traceability of driver to supervisor handover
- Finance struggles to verify supplier invoices at month-end, takes 3+ days
- Manual Excel data entry consumes 65+ hours per month
- Compliance risk for government contract audits

2.2 Goals

- **Zero lost D.O.s** — every delivery captured at source
- **Real-time visibility** — supervisors see D.O.s within minutes, not weeks
- **Automated reconciliation** — supplier invoices matched against D.O. records automatically
- **Tamper-proof audit trail** — GPS, timestamp, photo for every record
- **Eliminate manual data entry** — AI reads D.O. and fills the form

2.3 Out of Scope

- Multi-tenant SaaS (single-tenant only)
- Mobile native apps (PWA / responsive web only)
- Integration with existing accounting system (CSV/Excel export only)
- Driver payroll or HR features
- Vehicle GPS tracking (existing system handles this)

3. User Roles

Role	Count	Primary Device	Key Actions
Driver	~28	Mobile phone	Capture D.O. photo, submit form

Supervisor	~5	Mobile + desktop	Verify/flag D.O.s, view fleet activity
Finance	1–2	Desktop	Match invoices, export reports, audit trail
Admin	1	Desktop	Manage users, vehicles, suppliers, projects

4. Functional Requirements

4.1 Authentication & Authorization

FR-AUTH-1: Phone-number-based login for drivers (SMS OTP) **FR-AUTH-2:** Email + password login for supervisors, finance, admin **FR-AUTH-3:** Role-based access control (RBAC):

- Driver: can only see/submit own D.O.s
- Supervisor: can see all D.O.s, verify/flag
- Finance: can see all verified D.O.s, export, view invoices
- Admin: full access including user/master data management

FR-AUTH-4: Session timeout after 7 days of inactivity (drivers), 24 hours (admin/finance) **FR-AUTH-5:** Password reset via email

4.2 Driver Module (Mobile-First)

FR-DRV-1: Camera capture of D.O. photo

- Use device camera via
- Or HTML5 MediaStream API for in-app camera
- Compress to max 2MB, 1920px longest edge before upload

FR-DRV-2: AI auto-fill from photo (see AI Integration section)

- Extract: D.O. number, vehicle plate, supplier, quantity, material type, location, date
- Display extracted fields with confidence indicator
- All fields must be editable by driver before submission

FR-DRV-3: Manual form fields (with dropdowns):

- Vehicle plate (auto-filtered to driver's assigned vehicles)
- Supplier(s)
- Material type (TON / TIN / DRUM)
- Project code (master list, ~13 codes)
- Supervisor (master list, ~5 names)
- D.O. number (text)
- Quantity (decimal number)
- Location (text, optional)
- Remarks (text, optional)

FR-DRV-4: Auto-capture metadata:

- GPS coordinates (latitude, longitude) via Geolocation API
- Timestamp (server-side, not client)
- Driver ID (from session)

FR-DRV-5: Duplicate D.O. number detection

- Real-time check as driver types
- Show warning if D.O. number already exists in system
- Allow override with reason (in case of legitimate duplicate)

FR-DRV-6: "Today's Uploads" view

- List of D.O.s submitted by current driver today

- Status badge: Pending / Verified / Flagged
- Tap to view full details

FR-DRV-7: Offline support (Phase 2)

- Queue uploads when offline
- Auto-submit when connection restored
- Show pending sync count

4.3 Supervisor Module

FR-SUP-1: Dashboard with KPIs (today's view)

- Total D.O.s today
- Verified count
- Pending verification count
- Flagged count
- Total tonnage by type

FR-SUP-2: Daily D.O. table

- Sortable by time, vehicle, driver, supplier
- Filterable by project, supplier, status
- Each row: D.O. number, vehicle, driver, supplier, qty, type, GPS link, photo thumbnail, status, action buttons

FR-SUP-3: Verify action

- One-tap verify (changes status from Pending → Verified)
- Records verifier user ID and timestamp

FR-SUP-4: Flag action

- Opens modal to select reason (Missing info / Wrong vehicle / Duplicate / Quantity mismatch / Other)
- Optional notes field
- Records flagger user ID, timestamp, reason

FR-SUP-5: Photo viewer

- Tap thumbnail to open full-screen photo
- Pinch-to-zoom on mobile
- Show all captured metadata alongside photo

FR-SUP-6: Missing D.O. alerts

- Compare deployed vehicles (from existing schedule) vs. uploaded D.O.s
- Highlight vehicles with no upload by configurable cutoff time (default 6pm)

FR-SUP-7: Driver compliance scorecard

- Monthly view of each driver's upload count, verified rate, flag rate
- Color-coded: Green (90%+) / Amber (70-89%) / Red (<70%)

FR-SUP-8: Natural language search (AI-powered, see Section 8)

- Free-text input: "Show me all Highway D.O.s over 10 tons this week"
- Returns formatted answer with relevant D.O. records

4.4 Finance Module

FR-FIN-1: Monthly reconciliation dashboard

- Month selector
- Total D.O.s, total amount, total tonnage by month
- Breakdown by supplier (cards with match status)

FR-FIN-2: Supplier invoice upload

- Upload supplier invoice PDF
- AI extracts line items (D.O. numbers, quantities, amounts)
- Auto-match against verified D.O. records
- Flag discrepancies (missing D.O.s, quantity mismatches, price mismatches)

FR-FIN-3: Detailed D.O. table (verified only)

- All columns matching the existing Excel template format:

No, Date, D.O. No, Vehicle, Location, Net Weight, Type, Project, Supervisor, Remarks

- Filterable by supplier, project, type, date range

FR-FIN-4: Export to Excel

- Match the exact format of client's existing "Purchase Premix List" template
- One file per month, downloadable as .xlsx
- Include all verified D.O.s for the period

FR-FIN-5: Audit trail

- Log every action: upload, verify, flag, export
- Filterable by user, action type, date range
- Immutable records

FR-FIN-6: Discrepancy report

- List of flagged D.O.s with reasons
- Invoice vs. D.O. mismatches with dollar amounts
- Exportable to Excel

4.5 Admin Module

FR-ADM-1: User management

- CRUD for drivers, supervisors, finance users
- Assign vehicles to drivers
- Reset passwords / deactivate accounts

FR-ADM-2: Master data management

- Vehicles (plate number, type, status)
- Suppliers (name, address, default unit price by material type)
- Projects (code, name, manager)
- Material types (TON / TIN / DRUM with default unit prices)

FR-ADM-3: System configuration

- AI provider settings (API key, model)
- Cutoff time for missing D.O. alerts
- Email notification settings

FR-ADM-4: Bulk import

- Import vehicles, drivers, projects from CSV
- Initial data load from client's existing Excel files

5. Non-Functional Requirements

5.1 Performance

- AI extraction: <3 seconds per photo (95th percentile)
- Page load: <2 seconds on 4G mobile
- Database queries: <500ms for list views with up to 10,000 records
- Photo upload: <5 seconds for 2MB image on 4G

5.2 Scalability

- Support 100 concurrent users (initial)
- Handle 10,000 D.O.s per month
- Scale horizontally via Vercel Functions

5.3 Reliability

- 99.5% uptime (excluding planned maintenance)
- Daily automated database backups (retain 30 days)
- Graceful degradation if AI service is down (manual entry still works)

5.4 Security

- HTTPS only
- All photos and data encrypted at rest
- Row-Level Security (RLS) in Supabase
- API keys stored in environment variables only
- OWASP Top 10 compliance
- Rate limiting on AI endpoints (prevent abuse)

5.5 Compliance

- Singapore PDPA (Personal Data Protection Act) compliant
- Data retention: 7 years (matches government contract audit requirements)
- Audit log immutable
- Photo metadata preserved (EXIF stripping not allowed for legal records)

5.6 Browser Support

- Chrome / Safari / Edge (latest 2 versions)
- iOS Safari 15+
- Android Chrome 100+

6.0 Working Prototype Reference

Driver Upload

Supervisor

Finance

Snap a photo of any D.O. — AI reads and fills the form automatically

Upload Delivery Order

Tap to capture D.O. photo

Timestamp: --GPS: --

VEHICLE PLATE *

Select vehicle...

D.O. NUMBER *

e.g. D8080507923

MATERIAL TYPE *

Select type...

PROJECT CODE *

Select project...

DRIVER NAME

Auto-filled from vehicle

SUPPLIER *

Select supplier...

QUANTITY / NET WEIGHT *

e.g. 10.5

SUPERVISOR *

Select supervisor...

Driver Upload

Supervisor

Finance

Finance Reconciliation

Month: March 2026

TOTAL D.O.S
262

VERIFIED
209

TOTAL AMOUNT
\$41,153.11

TOTAL QTY (TONS)
706.9

PENDING VERIFY
40

FLAGGED
13

Export to Excel (CSV)

Print Report

Supplier Name

D.O.S
209

D.O. TOTAL
\$41,153.11

TON: 641.9 • DRUM: 1.0 • TIN: 64.0

TOTAL QTY
706.9

INVOICE AMT
\$41,153.289

Matched

D.O. Records — Verified

All Suppliers

All Projects

All Types

NO.	DATE	D.O. NO	VEHICLE	LOCATION	NET WT	TYPE	UNIT \$	AMOUNT	PROJECT	SUPERVISOR	STATUS
1						TON	\$56.00	\$112.00			Verified
2						TON	\$56.00	\$56.00			Verified

Driver Upload

Supervisor

Finance

Supervisor Dashboard

Date: 18/03/2026

Ask AI: "Show all D.O.s over 10 tons this week" or "Which drivers have the most flagged entries?"

Ask AI

Try: "total spending on premix in March" • "drivers with pending D.O.s" • "compare Highway vs Jin Quan"

TOTAL D.O.S
20

VERIFIED
19

PENDING
1

FLAGGED
0

Missing D.O. Uploads (21 vehicles)

XD3589T

XE8933U

YP1528H

Delivery Orders

All Projects

All Suppliers

All Status

#	TIME	D.O. NO	VEHICLE	DRIVER	SUPPLIER	MATERIAL	QTY	PROJECT	GPS	STATUS	ACTIONS
1						2 TON	2		View	verified	
2						1 TIN	1		View	verified	

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7. Recommended Stack

Builders may propose alternatives with justification. The following stack is preferred:

Layer	Technology	Rationale
Frontend	Next.js 15 (App Router) + TypeScript	Modern, performant, fits Vercel ecosystem
Styling	Tailwind CSS + shadcn/ui	Fast development, consistent UI
Backend	Next.js Route Handlers	Single deployment, serverless
Database	Supabase (Postgres)	Built-in auth, RLS, realtime
Auth	Supabase Auth	Phone OTP for drivers, email for staff
Photo Storage	Cloudinary	Already provisioned, existing preset
AI Provider	Google Gemini 2.5 Flash	Best OCR for documents, cost-effective
Hosting	Vercel	Single-click deploys, edge network
Monitoring	Sentry + Vercel Analytics	Error tracking and performance

8. AI Integration Summary

8.1 D.O. Photo Extraction

Endpoint: `POST /api/ai/extract` **Input:** Base64 image (max 2MB) or Cloudinary URL **Output:** Structured JSON with extracted fields **Provider:** Google Gemini 2.5 Flash via REST API (Google AI Studio)

- API key in `GEMINI_API_KEY` env var
- Max retries: 2 with exponential backoff
- Timeout: 10 seconds

Error handling:

- If JSON parsing fails → return null fields, prompt user to fill manually
- If API fails → fallback to manual entry (do not block submission)
- Log all extraction attempts to `audit_log` for monitoring accuracy

8.2 Supplier Invoice Extraction

Endpoint: `POST /api/ai/extract-invoice` **Input:** PDF file (max 10MB) **Output:** Array of line items + totals **Process:**

1. Convert PDF pages to images
2. Send to Gemini with prompt to extract D.O. references and amounts
3. Match extracted D.O. numbers against `delivery_orders` table
4. Compute discrepancies (missing D.O.s, qty mismatches, price mismatches)
5. Store in `supplier_invoices` table

8.3 Natural Language Search

Endpoint: `POST /api/ai/search` **Input:** Free-text query string **Output:** Formatted HTML answer **Process:**

1. Build a data summary (totals by supplier/driver/project/type) — see demo for working implementation
2. Send to Gemini with the query + data summary
3. Return AI-generated answer

Cost estimate:

- D.O. extraction: ~\$0.003 per call × 262/month = **\$0.79/month**
- NL search: ~\$0.005 per call × 100/month = **\$0.50/month**
- Invoice extraction: ~\$0.02 per PDF × 30/month = **\$0.60/month**
- Total: **~\$2/month** (negligible)

9. UI/UX Specs

9.1 Brand Guidelines

- **Primary:** Navy #1a3a5c
- **Accent:** Brown #8B5E3C
- **Success:** Green #22c55e
- **Warning:** Amber #f59e0b
- **Danger:** Red #ef4444
- **Background:** #f5f7fa
- **Font:** System font stack (-apple-system, BlinkMacSystemFont, 'Segoe UI', Roboto)

9.2 Additional Info

- Component styling (cards, buttons, badges, tables)
- Form layouts
- Color usage and status indicators
- Mobile responsive behavior
- Modal and dialog patterns

Developer should match the visual design exactly while implementing it properly in React/Tailwind.

9.3 Mobile-First Requirements

- Driver view must work flawlessly on iPhone SE (375px) and up
- Touch targets minimum 44x44px
- Forms vertical-stack on mobile
- Camera button prominently placed (thumb-reachable)
- Offline indicator when network drops

10. Testing Requirements

- Minimum 80% line coverage on backend code
- All API routes must have tests
- E2E tests for critical paths: driver upload, supervisor verify, finance export
- Test stack: Vitest (unit/integration), Playwright (E2E)

11. Third Party Integrations

The solution is expected to integrate with the following services:

- Google AI Studio (Gemini API)
- Cloudinary (image storage)
- Supabase (database and authentication)
- Vercel (hosting)
- Sentry (Error monitoring)
- Optional: SMS OTP for driver login (via Twilio or native auth provider)

12. Deployment & DevOps

The application is expected to be deployed on a modern cloud stack (e.g. Vercel or equivalent).

Basic expectations:

- Separate staging and production environments
- Simple CI/CD setup (e.g. auto deploy on merge)
- Basic error monitoring (e.g. Sentry or equivalent)
- Performance monitoring (e.g. built-in analytics)

Full deployment setup and environment configuration can be defined during implementation.

13. Acceptance Criteria

1. All Functional Requirements implemented and tested
2. AI extraction succeeds on >90% of D.O. photos (sample: 50 real photos provided)
3. Excel export matches existing template format byte-for-byte
4. Mobile driver flow completes in <30 seconds end-to-end (snap → submit)
5. Supervisor can verify 50 D.O.s in <5 minutes
6. Finance can reconcile a full month in <30 minutes
7. Lighthouse score: >90 on mobile (Performance, Accessibility, SEO)
8. Test coverage >80% on backend
9. All 28 drivers, 5 supervisors, and 1 finance user can log in and use the system
10. Production environment monitored with Sentry, no P0/P1 errors for 7 days

PART B: YOUR SUBMISSION

What We Need From You

Submission Process: Proposals will be reviewed in two stages to respect builders' time and streamline the evaluation process.

Stage 1 Initial Submission (Invited Builders)

Builders are invited to provide a high-level indicative response for (quotation range, timeline, team) to support early alignment.

What to Submit (Initial)

1. Indicative fixed-price quotation (range is acceptable)
2. High-level timeline
3. Proposed team composition (roles + experience)
4. Stack confirmation (or alternative with brief justification)
5. Any key assumptions or clarification questions

Timeline: Please submit within 2–3 business days.

Stage 2 Detailed Proposal (Selected Builders only):

Selected builders will be invited to submit a full proposal, including detailed cost breakdown, payment milestones, assumptions, and post-launch support model. Details should include:

1. Fixed-price quotation for the full scope with payment milestones
2. Timeline estimate with milestone breakdown and delivery dates
3. Team composition (developers, designer, QA) including allocation and optional rates
4. Assumptions and exclusions
5. Stack confirmation or proposed alternative with justification
6. [Optional / Separate] Post-launch support model (warranty + maintenance + SLA)
7. Any additional questions or clarifications

Timeline: To be confirmed once selection is complete (typically within 3-5 business days).

What to Submit (Selected Builders only):

1. Quotation

Please provide your fixed-price quotation broken down by deliverable. Quote in SGD. If quoting in USD, include the SGD equivalent at time of submission.

Deliverable	Key Scope	Quote (SGD)
Foundation & Auth	DB schema, auth, base UI, RBAC	\$
Driver Capture & AI Extraction	Camera capture, Gemini OCR, form submit, duplicate detection	\$
Supervisor Dashboard	Verify/flag, KPIs, missing D.O. alerts, compliance scorecard	\$
Finance & Reconciliation	Invoice PDF upload, AI matching, discrepancy report, Excel export	\$
Admin, Master Data & NL Search	User management, master data, bulk import, AI search	\$
Testing, UAT & Production Deploy	E2E tests, UAT with client, go-live, handover	\$
Total Fixed-Price Build Cost		\$

Payment milestone split (e.g. 40% on signing / 30% at mid-point / 30% at go-live): _____

Currency SGD / USD (please specify, include SGD equivalent if quoting in USD): _____

Third-party costs passed to client (if any): _____

2. Timeline & Milestones

Please propose your own milestone breakdown with demo checkpoints after each phase:

Deliverable	Suggested Deliverables	Your Estimate
Phase 1: Foundation & Auth	Database schema, auth, base UI	___ weeks
Phase 2: Driver Capture & AI Extraction	Photo capture, AI extraction, form submission	___ weeks
Phase 3: Supervisor Dashboard	Dashboard, verify/flag, missing alerts	___ weeks
Phase 4: Finance & Reconciliation	Reconciliation, invoice matching, Excel export	___ weeks
Phase 5: Admin, Master Data & NL Search	User management, master data, AI search	___ weeks
Phase 6: Testing, UAT & Production Deploy	E2E tests, UAT, go-live	___ weeks
Total		___ weeks

3. Team Composition (developers, designer, QA)

Provide your proposed team:

- Roles (e.g. frontend dev, backend dev, QA, designer)
- Years of experience per role
- Relevant project experience (if any)
- Estimated allocation (full-time / part-time)
- Hourly or daily rates (optional)

4. Assumptions & Exclusions

Kindly provide in bullet points:

<Insert assumptions / exclusions here>

5. Stack confirmation

To confirm stack or propose alternative with justification.

<Insert stack confirmation / proposed alternatives here>

6. Post-Launch Support & Maintenance

The build is fixed-price. For completeness, please provide a clear breakdown of your post-launch support model if any. We will use this to compare total cost of ownership across proposals.

6.1 Warranty Period

- What's the period of free bug fixes are included post-launch before the retainer begins?
- What is covered during the warranty period (bug fixes only, or includes minor changes)?

6.2 Retainer post launch

If any, please specify what is included and excluded:

Item	Your Response
Monthly retainer fee	
What is included	e.g. bug fixes, infra monitoring, minor UI updates, model retraining
What is excluded	e.g. new features, additional users, major architecture changes
Infrastructure costs post-handover	Who pays for Supabase, Vercel, Cloudinary, Sentry etc. after go-live? Provide monthly estimate.
Support channel	e.g. email ticketing, WhatsApp, dedicated Slack channel

6.3 SLA Requirements

Please provide your proposed SLA for the production environment:

Critical (system down / no D.O. capture)

Response time: ___ hrs

Resolution time: ___ hrs

High (dashboard / sync issues)

- Response time: ___ hrs

- Resolution time: ___ hrs

Medium (UI bugs / minor issues)

- Response time: ___ hrs/days

- Resolution time: ___ days

Uptime commitment: ___ %

Weekend / public holiday coverage:

(System operates 6 days/week including Saturdays — please state coverage)

Scheduled maintenance notice:

Minimum advance notice: ___ hrs

7.Open Questions

Please address these in your proposal or flag for discussion:

- Driver authentication: Supabase Phone Auth (free) or Twilio (more reliable, ~\$0.01/SMS)?
- Domain: Custom domain or *.vercel.app?
- Notifications: WhatsApp alerts for flagged D.O.s? (requires WhatsApp Business API)
- Language support: Mandarin/Tamil needed for any drivers?
- Historical data migration: Import 6 months of existing D.O.s from Excel?
- Hardware: Will drivers use personal phones or company-issued devices?

PRESENTED TO YOU BY

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We look forward to the potential partnership together!